THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The idea of changing and observing step by step to answer a question is called: 1

(a) random guessing **(b)** systematic investigation
(c) memorising **(d)** copying

2. The Shaka era calendar shown in the chapter is an example of a: 1

(a) circuit diagram **(b)** recipe
(c) calendar based on sky observations **(d)** map

3. The instrument used to see microorganisms is: 1

(a) telescope **(b)** thermometer **(c)** spring balance **(d)** microscope

4. Scurvy is caused by lack of: 1

(a) iron **(b)** vitamin A **(c)** vitamin C **(d)** iodine

5. To avoid antibiotic resistance, we should: 1

(a) share leftover tablets **(b)** take antibiotics for cold
(c) take antibiotics only when prescribed **(d)** stop the dose when feeling better

6. Earth's magnetic field is produced by: 1

(a) volcanoes only **(b)** ocean waves
(c) movement of liquid iron in the core **(d)** the Moon

7. In which industry is a high-temperature electric furnace used? 1

(a) curd making **(b)** paper folding
(c) fruit packing **(d)** steel manufacturing

8. Friction is greater on a: 1

(a) smooth surface **(b)** ice **(c)** rough surface **(d)** polished glass

9. A knife cuts better with a sharp edge because: 1

(a) smaller area gives more pressure **(b)** it is magnetic
(c) it is heavier **(d)** larger area gives more pressure

10. The wall of seawater pushed by cyclone winds is called: 1

(a) tide **(b)** flood plain **(c)** storm surge **(d)** tsunami

11. When water boils, it changes into: 1

(a) solid **(b)** ice
(c) water vapour (gas) **(d)** salt

12. Compounds have elements combined in a: 1

(a) zero ratio **(b)** varying ratio **(c)** random ratio **(d)** fixed ratio

13. Compounds made only of non-metals include: 1

(a) brass and bronze **(b)** iron and gold

(c) salt and rust only **(d)** carbon dioxide and water

14. A solution can be turned from saturated to unsaturated at the same temperature by: 1

(a) adding more solvent **(b)** shaking it

(c) adding more solute **(d)** cooling it

15. When an object is far from a concave mirror, the image is: 1

(a) erect **(b)** invisible **(c)** inverted **(d)** always enlarged

16. After purnima, the bright part of the Moon: 1

(a) increases **(b)** stays same

(c) decreases **(d)** disappears at once

17. In the chain grass → grasshopper → frog → snake, the secondary consumer is: 1

(a) grasshopper **(b)** frog **(c)** snake **(d)** grass

18. A forest is an example of a: 1

(a) aquatic ecosystem **(b)** freshwater ecosystem

(c) terrestrial ecosystem **(d)** marine ecosystem

19. Venus is hotter than Mercury due to: 1

(a) ozone layer **(b)** distance from Sun

(c) no atmosphere **(d)** a strong greenhouse effect

20. Ozone is a form of oxygen with: 1

(a) one oxygen atom **(b)** two oxygen atoms

(c) three oxygen atoms **(d)** four oxygen atoms

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is the role of Lactobacillus? 2

22. What is immunity? 2

23. Who discovered the magnetic effect of electric current and in which year? 2

24. Explain how a rubber sucker sticks to a wall. 2

25. How can a mixture of iron filings and sulfur be separated? 2

26. Write two units of density. 2

27. What are trophic levels? 2

28. Name two features of Earth that make it suitable for life. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Why are ecosystems important enough to be studied in this book? 3

30. Draw a labelled diagram of a spring balance and name its parts. 3

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31. State why the boiling of water at 100 °C is not at room temperature. 3

32. List three differences between a concave mirror and a concave lens. 3

33. Calculate the gap in days between 12 lunar months and a solar year of 365 days. How many years will give one extra month approximately? 3

34. How does ozone depletion harm life? 3

35. Name three metal pairs that can be used in voltaic cells and state which usually acts as the positive electrode. 3

OR

Explain how to make a simple electromagnet. 3

36. State three benefits of healthy ecosystems for farms. 3

OR

Explain why arrows in a food chain point from the eaten organism to the eater. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Explain with examples how scientific heritage of India (Ayurveda, variolation) contributed to health ideas and what modern science adds. 5

OR

Write an essay on why prevention is better than cure, using at least five points. 5

38. Calculate pressure for 2 situations: a 40 N brick on its 0.1 m² face and on its 0.05 m² face. Explain the difference. 5

OR

Write an account of how air pressure and wind are connected, including breeze, storm and cyclone. 5

39. Describe how a solute behaves when a saturated solution is cooled and show it with a crystallisation example. 5

OR

Calculate the density of three liquids with the following (mass, volume): A (50 g, 50 mL), B (80 g, 100 mL), C (120 g, 100 mL). Which will float on water? 5

40. Draw and explain a grassland food web containing grass, grasshopper, mouse, frog, snake and hawk, and say what happens if snakes are removed. 5

OR

Write about Vrikshayurveda and traditional approaches to soil health compared with modern chemical farming. 5

END OF QUESTION PAPER — Total: 80 Marks

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